



Environmental aspects of material efficiency versus carbon storage in timber buildings

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Received: 23 February 2017 / Published online: 6 December 2017
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Abstract

This article highlights the interdependencies of carbon storage and material efficiency in timber buildings. These elements appear contradictory in regard to environmental factors. Following a general literature review of environmental aspects of timber buildings and their components, building elements are analyzed in regard to their relevance for both issues. Particular advantages for material efficiency and carbon storage in regard to specific building elements are discussed. The first section summarizes what currently qualifies as the state of art in environmental standards of timber buildings on building level, looking at influence on the sector and on the competition on wooden material for material or energetic use. Calculations on carbon storage of exemplary timber buildings are shown in the following section. A factor for material efficiency of wooden use is hereby introduced. The interdependencies of carbon storage and material efficiency are discussed on the level of building elements and according to necessary requirements. Results show that there is no “best” timber construction that fulfills all requirements, as each building and with that each specific construction needs to fulfill varying circumstances. It is thus demonstrated that it is not enough to only focus on one of the issues. It is important to keep these interconnections in mind for climate discussions in regard to timber buildings and to further optimize timber constructions and buildings to meet both targets.

1 Introduction

As a result of the Paris Climate Conference 2015—COP21—climate protection will increasingly focus on closing carbon cycles and reductions of greenhouse gas (GHG) emissions (COP21 2016). The building sector was identified as a major player for reducing GHG emissions in Europe. For instance, the sector accounts for about 40% of total primary energy consumption in the European Union, which has severe environmental impacts (EC 2007). In the context of sustainable development, a reduction of environmental impacts during a building’s life cycle is a significant target. Since 1990, quantifying and understanding environmental impacts has been applied to building sustainability assessment (Fava 2006) and numerous studies related to building life cycle assessment (LCA) have been undertaken internationally (e.g. Passer et al. 2014; Kuittinen et al. 2013; Graubner et al. 2006; Kaufmann and König 2011). In recent years

and with better energy efficiency in the use phase, the focus has begun to shift to used materials and their environmental impact (Weissenberger et al. 2014). To reach climate protection targets the building sector and especially timber as building material can play an additional role in reducing global warming potential (GWP). Here, the dual focus is on the issue of carbon footprint, on the one hand, and resource efficiency of building material (EC 2011), on the other hand (Weidema et al. 2008; Wiedmann 2009). In this article, the term “material efficiency” is used instead of resource efficiency, since the analyses focus only on the material use of wood. Timber buildings are defined as buildings with their load-bearing structure in timber. The focus is on buildings from cross laminated timber (CLT) and timber frame constructions, which are the most common types in modern timber structure.

This article highlights the interdependencies of carbon storage and material efficiency in timber buildings. The elements material efficiency and carbon storage appear contradictory in regard to environmental factors and need to be examined in greater detail. On the one hand, carbon storage capacities contribute to the forest’s carbon sink (Rüter 2011), on the other hand material should be applied

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in a useful and efficient way to fulfil the demand. However, high carbon storages require high input of wooden material. Timber buildings and with that their relevant building elements will thus be analyzed in regard to their relevance for both issues. Particular advantages for material efficiency and carbon storage in regard to specific building elements will be discussed. It is important to keep these interconnections in mind for climate discussions in regard to timber buildings.

1.1 Aim of the article

The observations and ideas in the article are based on the outcome of a research project on linking environmental requirements to the selling of building plots on an exemplary site (Hafner 2017; Hafner et al. 2017). In that project, timber as building material was introduced through requirements regarding the amount of renewable material used per m² gross external area (GEA). Based on the amount of carbon storage, subsidies were granted by the municipal government. During this project, the question arose of what is of greater importance for optimization: material efficient use of wood or large carbon storage influenced by linking environmental requirements of building plots to amount of renewable material?

1.1.1 Resource and material efficiency in the context of building with wood

Both the material use of wood and the energetic use of wood and bio-based materials (for example, bio-plastics) are of great importance for realizing both climate protection goals and energy transition. Such competition regarding wooden material greatly influences future wood use. The still rising amount of wooden material for energetic use will influence how wood and forests are dealt with. A variety of research has been done to investigate these correlations. Dieter and Seintsch (2012) stated that in the medium term significant shortages of coniferous round wood are expected in Germany. Trømborg et al. (2013) analyzed wood pellet production for various countries, showing that future pellet production will be based on round wood and wet sawdust. Such feedstock is also in demand by wood-based industries. Mantau (2012) calculated timber stocks and use of timber in Germany. For the limited market in Germany, the outcome of Mantau's study shows that the wood consumption (in m³) for energetic use (heat energy) already exceeded material use of wood in 2010.

The quantities of reclaimed wood have also been quantified. Wood cascading could support achieving higher quantities for wood use. In Höglmeier et al. (2013), wood cascading is also seen as an option to increase the availability of wood. On the basis of regional observations from southeastern Germany, the resulting streams of recovered

wood from the building stock were calculated to locate environmentally beneficial cascading options for these resources. As an outcome, around 50% of the recovered wood was in suitable condition for reuse or high-value secondary application. A better management of renewable resources supports the material supply of the wood sector to ensure a long-term availability of solid wood products at reasonable prices. Additionally Höglmeier found that in comparing LCA of primary wood and recovered wood the cascading systems create fewer environmental impacts, if system expansion is based on wood energy. To maximize the benefits and minimize the associated environmental impacts, cascading proves to be a preferable option of utilizing untreated waste wood (Höglmeier et al. 2014). In Höglmeier et al. (2015), LCA calculations on cascading options for wooden products showed that cascading leads to saving up to 14% of the annual primary wood supply in a regional study area in Germany.

Hafner et al. (2013) discussed a realistic estimation of theoretical scenarios for end-of-life of timber from buildings and their influence on planning processes. Wide span structures with large timber construction are especially interesting for reuse, whereas other wooden parts from old houses can only be used for cascading and need intensive screening of preserving agents beforehand (Hafner et al. 2013).

Material use of timber, ideas around material efficiency need closer inspection so that wood consumption is adjusted to constructive needs. Therefore, the material efficient use of timber material in building must be looked at closer in terms of long-term availability of wooden material. The actual expertise of the Scientific Advisory Board on Forest Policy (BMEL 2016) suggests therefore, to increase the longevity of wood products and to promote their cascade use.

Resource efficiency has been an issue in the scientific community for a long time and was introduced in the political agenda in European Community since 2011 (EC 2011). "The Roadmap to a Resource Efficient Europe" proposes pathways for increasing resource productivity, for example by including decoupling economic growth from primary resource use. Worldwide depletion of natural resources and increasing global demand have pushed the efficient use of natural resources into the political forefront (EC 2016). The EU Directives as well as the national laws in many European countries aim at higher rate of reuse and recycling. The basic principle in the European Waste Management Directive is that materials should be primarily recovered for secondary use, and only as a secondary option, should they be utilized as energy.

In general, resource efficiency is defined by using the earth's limited resources in a sustainable manner while minimizing impacts on the environment (EC 2011). In other words, resource efficiency is the relation of effect (product or service) to the input of natural, mainly abiotic, resources.

Currently, there is an ongoing discussion in Germany on resource efficiency and its measures and influences, although discussions mainly focus on abiotic resources and not on biotic resources like wood (BBR 2016). For the article only the aspect of material efficient use of wooden products is looked at—not the resource efficiency in general building sector.

The term material efficiency is used as expression for the efficient use of wood in buildings. In this context “material efficiency” is defined as the environmental effects for the building, here expressed in GWP per functional unit of the building in reference to the input of wooden material of the same functional unit. In the literature reviewed, the only study that focused on wooden (light-weight) material in reference to resource efficiency in an LCA by combination of GWP in lifecycle and potential benefits is that by Knauf (2015). The ideas proposed there do not help to define material efficiency in wooden buildings, as they are explained on the basis of a single wooden product only and not refer to a whole building.

1.1.2 Carbon storage

According to the LCA standard EN 16485 carbon storage is defined as biogenic carbon, which is stored over a fixed period of time. Hence, the carbon storage is calculated as neutral over the products life cycle, which means that after the end-of-life stage of a product its carbon storage is zero.

The underlying principle is that in the product stage of a wooden product (module A1 until A3) the carbon storage is indicated as GWP with a negative sign. In the end-of-life stage (module C) the stored carbon is discharged of the LCA system. That means that the longer a product is materially used, the longer the biogenic carbon is stored in the product. Additionally the more wood is used, the higher the carbon storage is.

High carbon storage preservations of harvested and materially used wood contribute to the national CO₂ accounting since the implementation of the second period of the Kyoto Protocol (2013–2020) (Umweltbundesamt 2014).

1.1.3 Approach

For answering the question: *what is of higher importance for timber buildings in terms of climate benefits: material efficient use of wood or large carbon storage?*, this article is separated into two main parts in the methods section: Part I (Sect. 2.1) analyses different contemporary building types (timber, hybrid, mineral). The focus is on the timber buildings with typical constructions types (CLT, timber frame). Mineral buildings are shown for comparison with the timber buildings. The comparison of LCA is made on building level.

The account of the carbon storage is part of the LCA study and expressed via the indicator GWP in accordance with EN 15978, (EN 15804 2014) and is also demonstrated in this section. Based on the LCA results a factor is introduced to show the efficient use of wooden material in buildings in terms of GHG emissions per wood used.

In part II (Sect. 2.2), the interdependencies of carbon storage and material efficiency are discussed on the level of building elements and according to necessary technical requirements. This discussion is not done on building level but on level of building elements, as various combinations are possible and assemble a timber building. The dependencies and connections of single structural elements in buildings are analyzed, highlighting the pros and cons of the constructions in terms of their environmental qualities. The question of higher material efficient use of wood or more carbon storage capacity is discussed for each element and various requirements are shown.

1.2 Building level—state of art of environmental standards and research results on product level

Various studies were conducted comparing conventional buildings and timber buildings to ascertain specific advantages and specificities of the latter.

Takano’s comparison resulted in high carbon storage capacity for timber constructions with large quantity of renewable primary energy bound in material (PERM) for erection stage of the building (module A) (Takano et al. 2014a). The advantages of timber constructions as a binder of primary energy are to a great extent balanced out over the whole life cycle (module A, B,C) and only result in smaller net energy benefits from material recycling (module D) (Takano et al. 2014b). Heeren et al. (2015) showed that in wooden buildings the median GHG emissions are noticeably lower, even when credits at end-of-life are excluded. Similar studies on the advantages of timber construction have been conducted by Dodoo et al. (2012). Kaufmann and König (2011), Ximenes and Grant (2013), König (2016) and Hafner (2014). Comparing timber buildings using different construction methods, the carbon implications in life cycle were shown. Hereby, CLT construction offers lowest life cycle GHG emissions (Dodoo et al. 2014). Kaufmann and König (2011) calculated LCA for various realized timber buildings and compared them to mineral buildings with the same functions. All buildings showed advantages for timber buildings regarding GWP and indicators for primary energy (Kaufmann and König 2011; König 2016). Ximenes and Grant (2013) showed that significantly lower GWP values were achieved by optimising the use of wood products for two common house designs in Sydney. Published works by Buyle et al. (2013) and Cabeza et al. (2014) also showed that buildings made with mainly renewable material offer lower

environmental impacts compared to buildings with mineral buildings materials.

These studies all differ in approach, system boundaries, database, and scope, and therefore cannot easily be compared. Additionally, they are in part not yet executed according to the actual standard for buildings in LCA (EN 15978 2012). They all, however, demonstrate that on the level of construction material in product stage (module A1 until A3), wooden material has advantages in terms of carbon storage capacity, therefore resulting in lower GHG emissions in the product stage (EN 16485 2014). They also reveal that calculations over the whole lifecycle show a slightly different picture. Here, the advantages are generated mainly through benefits outside system boundaries (module D). This is directly connected to the use of wooden material after the end of life status by cascading or use for energy substitution.

On building product level, additional research on reviewing and comparing LCA calculations for wooden building material was done by Wolf et al. (2016), stating that wooden building materials can be described as “low emission raw materials”, if sustainable forestry can be presumed. But existing standards (e.g. EN 16485) should be used to help comparability of data (Wolf et al. 2016). Achenbach and Rüter (2015) carried out LCA on average prefabricated timber buildings produced in Germany to determine the environmental impacts of the production and construction stages as well as its particular building elements according to standards. In depth calculations on environmental properties of wooden building material according to new standards EN 15978 and EN 15804 were undertaken by Rüter and Diederichs (2012), which are now integrated in the German dataset ökobau.dat (BBSR 2015). Fadai and Winter (2015) and Fadai et al. (2013) studied the combination of wooden material and other materials, such as lightweight wood particle concrete for structural construction or wood-based, multi-layer systems. These composite systems offer a number of interesting prospects in terms of resource efficient use of wood combined with structural performance and environmental profiles. Resource efficient use of wood here is seen as optimized use of materials according to their mechanical properties and the proposed use in buildings. In addition, Kain et al. (2015) researched on improving the use of tree bark as a building material. These studies aim at optimizing construction elements or singular layers with the use of renewable material in the context of increased environmental performance.

In general, all these findings show environmental benefits through building with wood in terms of the indicators of GWP and primary energy renewable for material use, and additionally in terms of carbon storage capacity. The studies differ in detail due to various system boundaries, datasets and the possible inclusion of biogenic carbon, as well as the calculation of benefits after end of life.

1.3 Building sector approach

As previously shown, timber used in buildings is according to EN 16485 a temporary (product) carbon storage that exists until the wooden building element is disposed.

From a LCA standard's viewpoint, the longer a wooden product is used materially within the building's life cycle, the longer the carbon storage can be preserved. Temporarily stored carbon can play an additional role in improving the effectiveness of a forest's carbon sink. With the second period of the Kyoto Protocol (2013–2020), the rules for reporting were adapted and the inclusion of forest management was made compulsory (Umweltbundesamt 2014). With this implementation, the material use of wood in the building sector is accounted for. Since then, temporary and dynamic changes in the carbon pool of harvested and used wood must be considered. According to article 3.4 of the Kyoto Protocol, deductions are applied both only on national level and for domestic wood use only. The quantification of expected climate impact deriving from rising material use of timber is therefore relevant on a national scale and can enhance the carbon sink of the forest sector. The amount of carbon is quantified on the basis of sawn timber, derived timber products, and paper (Rüter 2011).

In addition to carbon storage capacity, there is a substitution potential (material and energetic). Material substitution can be realized by replacing building materials made of non-renewable materials (such as plastics, metal, or concrete) with renewable material. Energetic substitution is realized by the use of renewable material instead of fossil fuels. In a meta-analysis, Sathre and O'Connor (2010) calculated a general substitution factor for material substitution in a comparison of data from various studies. This factor constitutes around 1.9 t CO₂ equivalent per m³ wood, when timber replaces concrete or masonry in buildings. Calculations in the authors' own research suggest slightly smaller results by applying actual calculation methods according to EN 15978 (ReB 2017).

In actual national accounting there are four effects of carbon capturing: (1) carbon pool of forest; (2) carbon pool of products; (3) material substitution; and (4) energetic substitution. Number (1) and (2) are accounted for in the national inventory report and declared to the climate secretary of UN. Number (3) and (4) are avoided emissions, which are not directly accounted for in the sector but may help to reduce emissions from other sectors (Heuer et al. 2016) when sustainable forestry can be presumed.

While the inclusion of substitution and temporary carbon storage capacity on the achievement of the climate protection goals can only be applied to the entire national-wide sector, it can be demonstrated on a building level. By choosing timber construction (timber frame, CLT, hybrid) for a single building, the amount of carbon storage can be

influenced. After the full lifecycle of a (wooden) building, the extracted renewable material can replace fossil fuels and generate an energetic substitution effect in module D (according to EN 15978). This means that on building level the carbon storage capacity can be shown. Through inclusion in national calculations they contribute to the quantification of carbon sink (Rüter 2011; ReB 2017) and are of interest in a wider context.

The aim of this article is to highlight the interrelations between material efficient building with wood, the environmental quality of carbon storage, and the relevant practical use for planning and building construction. In general, long-term and material efficient use of premium quality wood (such as laminated wood, plywood, timber frame construction) is necessary to ensure sustainable construction with wood. In the process of planning wooden construction, the deconstruction, reuse, and recycling of the products also need to be considered, always in accordance with the actual standards EN 15804 and EN 15978.

2 Materials and method

2.1 LCA and material efficiency factor on building level

Keeping in mind the advantages of timber in the building sector as explained in Sect. 1, closer research to analyze different timber buildings was undertaken. Timber buildings can range from only having external walls in timber frame construction (hybrid buildings) (Neubauer-Letsch et al. 2013) up to sole timber buildings where CLT is used as the load-bearing structure. The different constructions result in different amounts of used timber material, and therefore in different carbon storage and GWP.

Based on this question, LCA results from the research project “Greenhouse gas (GHG)—timber buildings” (ReB 2017) are evaluated. First, the underlying research project is briefly described. Results of LCA calculations are placed in the context of the climate effects of GWP and carbon storage as well as material efficiency. In Sect. 2.1.5 a factor for material efficient use of wooden material in buildings is introduced to relate environmental emission to the use of wooden material. Through that the different construction types of timber buildings are evaluated. Buildings with mineral constructions (concrete, brick, sand-lime brick) are exemplarily enlisted for comparison.

2.1.1 The research project “Greenhouse gas—timber buildings”

In the research project “GHG—timber buildings” substitution factors on building level were identified for new

residential timber buildings on the basis of actual standards in LCA calculations. Beside positive substitution potential of wooden products, the carbon storage for timber buildings was quantified for Germany to show possible consideration in the Kyoto Protocol. Based on actual market share of timber buildings the project showed the impact of changes in market share for use of wood until 2030 and their impact on GHG emissions. Obstacles in timber buildings, which opposed rising share of wood construction were investigated and solutions were developed. For this article only the LCA calculations on building level are used. The following sections show the frame conditions, the analyzed buildings and results of the LCA performance.

2.1.2 Frame conditions

The general frame conditions for these LCA study are briefly explained.

2.1.2.1 Software and database oekobau.dat LCA calculations were performed with the LCA-tool LEGEP (Ascona 2016) which is a tool especially for the building sector. The underlying database is the Oekobau.dat published by the Federal Ministry for the Environment, Nature Conservation, Building and Nuclear Safety (BBSR 2015). The oekobau.dat 2015 was adapted to the new standard EN 15804 and predominantly fulfils the requirements on LCA performance. Calculations are limited to German context.

2.1.2.2 System boundaries

• Modules

Calculations were performed according to the standards EN 15978, EN 15804 and EN 16485, see Fig. 1.

The calculation method and considered modules are geared to the German Assessment System for Sustainable Building (BNB) (BMUB 2012).

Within the system boundaries there are the modules A1–A3 (product stage), module B2 (maintenance such as protective coatings but without cleaning), module B4 (replacement) and the modules C3–C4 (waste processing and disposal). Module D (benefits and loads beyond the system boundary) is noted as supplementary information beyond the building life cycle, but is shown separately as required by the standard.

Module A4 (transport from manufacturing to the construction site), module A5 (construction-installation process), module B1 (use), module B2 (cleaning), B3 (repair), B5 (refurbishment), B7 (operational water use), C1 (deconstruction, demolition) and C2 (transport from construction site to waste processing/disposal) are not considered in building LCAs according to the calculation rules of the

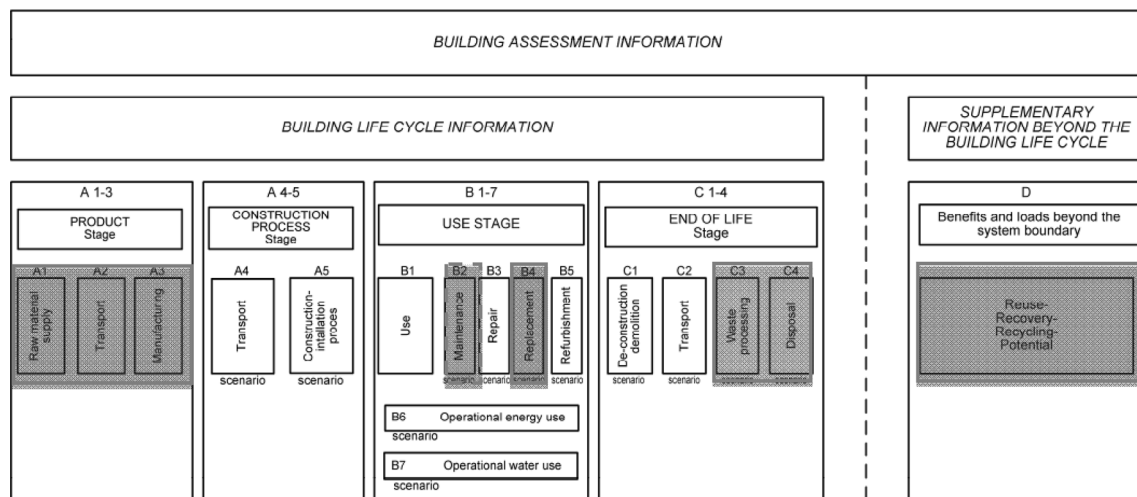


Fig. 1 Display of modular information for the different stages of the building assessment (EN 15978: 2012). The framed modules are part of the LCA, whereas module D is calculated separately

BNB system. (BMUB 2012). The non-consideration of these modules is attributed to the lack of data in the oekobau.dat. Furthermore, module B6 (operational energy use) is not included since the buildings to be compared are functionally equivalent and have an identical energy standard, determined according to Energy Conservation Act/EnEV rules (DIN 4108 and DIN 4701). For this reason the energy use is not relevant for the LCA results and was left out.

The LCAs were calculated over a life cycle of 50 years according to the German Assessment System for Sustainable Building (BMUB 2011).

- Product system

All buildings were calculated without basement to overcome the influence of basement on the results. Instead, all buildings were calculated with unified foundations. Outdoor facilities on the building site and finishing such as doors, flooring, windows, roof covering, façade cladding, stairs and technical building equipment are not accounted for in these LCA calculations, since they are independent of the buildings construction and will appear in all buildings. Finishing elements, regardless of which material, could be applied to all construction types and would change results noticeably if not consistently used. For this reason finishing elements are not part of the analysis.

The building construction includes foundation, external walls (without cladding), internal walls and pillars, ceiling, roof construction and balcony (if present). Insulation materials are also included, since the buildings have to be functionally equivalent for LCA comparison. This means that all buildings fulfill the legal requirements concerning

noise and fire protection, energetic standard and have the same qualitative and technical functionality. The energetic performance of the buildings at least complies with Energy Conservation Act/EnEV rules (EnEV 2009) or better, hence all analyzed buildings have an approximately identical energetic standard.

To sum up, the building LCAs include all construction elements as above-mentioned plus materials which are required for guaranteeing functional equivalence of the buildings (e.g. insulation).

2.1.3 Analyzed buildings

The analyzed buildings in this article are large residential buildings with 3 floors or more and from about 700 m² up to about 7000 m² GEA. Most of the buildings are realized objects located in Germany and Austria with construction years between 2006 and 2014. Various construction methods and materials (timber, mineral and hybrid constructions) were used, as shown in Table 1. The construction types are defined by the material of their load bearing elements. Hybrid buildings consist at least of a load-bearing concrete structure and a façade in timber frame construction (Neubauer-Letsch et al. 2013). The hybrid buildings in this study exceed these minimum requirements for hybrid buildings. Hy_1 partly has a load-bearing timber frame construction (external wall) with the rest of the building elements in massive construction. Hy_2 uses CLT and timber frame as part of the load-bearing walls; in addition the ceilings are of composite materials from concrete and wood.

Table 1 Analyzed buildings

Number	Building type	Building (external wall construction + insulation)	Floor area GEA (m ²)	Number of floors	Year of erection
T_1	Timber	CLT + mineral wool	2033	8	2011
T_2	Timber	CLT + mineral wool	723	4	2010
T_3	Timber	Timber frame + ETICS ^a	1394	6	2013
T_4	Timber	Timber frame + cellulose and mineral wool	6152	3	2006
T_5	Timber	Timber frame + mineral wool	2717	4	2013
T_6	Timber	Timber frame + mineral wool	1257	4	2011
Hy_1	Hybrid	Timber frame + mineral wool	1172	5	2013
Hy_2	Hybrid	CLT, timber frame, reinforced concrete, mineral wool, ETICS	3847	5 and 7	2013
M_1	Mineral	Reinforced concrete + ETICS	1394	6	2013
M_2	Mineral	Porous concrete	2717	4	2013
M_3	Mineral	Perforated brick + insulating plaster	723	4	2010
M_4	Mineral	Sand-lime brick + ETICS	1478	4	2010

^aExternal thermal insulation composite system

T timber building, *M* mineral building, *Hy* hybrid building

2.1.4 LCA results

Results of the LCA calculations for GWP (kg CO₂ eq/m² GEA) and for carbon storage (kg CO₂ eq/m² GEA) are shown in Fig. 2, split in module A, B and C. Carbon storage is presented through the indicator biogenic GWP. The LCA performance was verified with a critical review according to EN ISO 14044.

As the material is introduced in the building in module A, the biogenic carbon content is indicated as negative GWP value. It leaves the system in module C and therefore the same amount of biogenic GWP is present with positive sign. The total GWP for a building is the sum of fossil and biogenic GWP.

Timber buildings have higher biogenic carbon shares than hybrid or mineral buildings. Still, mineral buildings

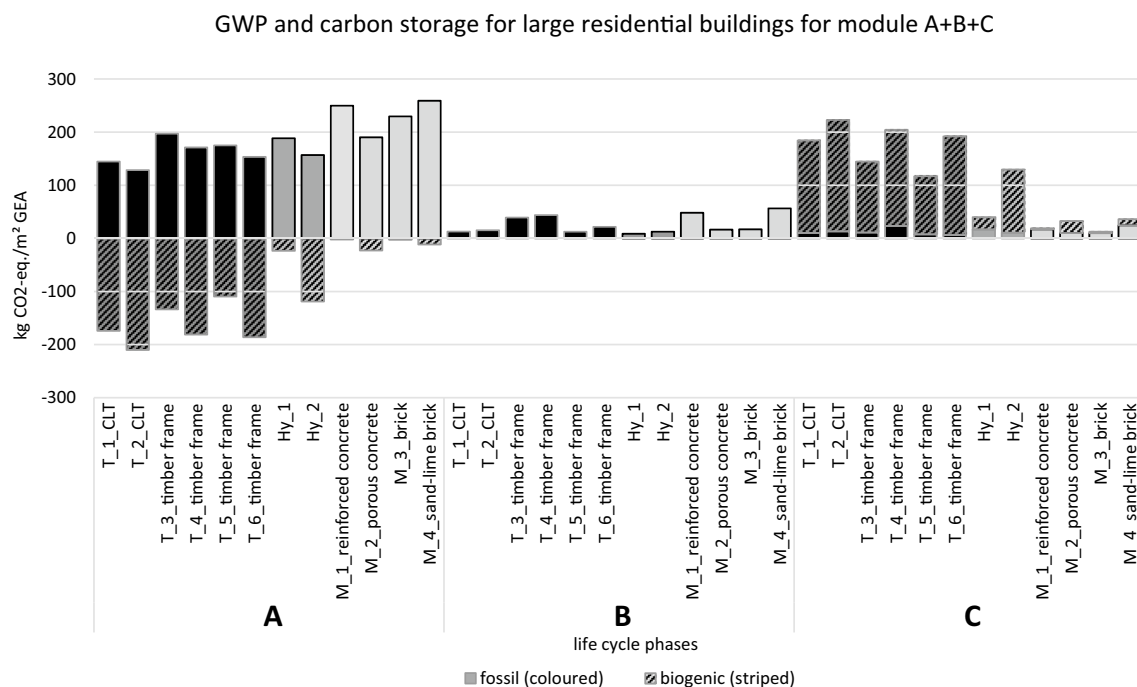


Fig. 2 GWP and carbon storage for the analyzed large residential buildings per m² GEA

also partly include timber elements (wooden attics, roofs) and therefore shares of biogenic GWP. The higher the share of wooden building products, the higher the biogenic GWP value. The biogenic share of the hybrid buildings varies since the amount of wooden material can diverge widely in hybrid constructions. Fossil GWP ranges from 128 to 197 kg CO₂ eq for timber buildings in module A and from 6 to 23 kg CO₂ eq for module C. Fossil GWP for mineral buildings is higher in all modules and ranges in module A from 157 to 259 kg CO₂ eq and in module C from 10 to 25 kg CO₂ Eq. In general, module A shows the highest emissions, followed by small emissions in module B and C. Emissions in module B result from replacement of ETICS, façade layer and general maintenance of building parts in the construction elements of the building. The height of biogenic carbon varies due to construction from −210 to −109 kg CO₂ eq for timber buildings. In general, the highest amount can be found in buildings made with CLT for floors and walls. But for buildings constructed with floors from CLT, timber frame exterior wall, insulation from renewable material and additional building parts in timber (building T_4) high values can also be achieved.

Further carbon storage capacity can be obtained by using timber material for the interior finishing like doors, windows, flooring, etc. These elements can be installed independently from the building construction and are not incorporated here.

Timber buildings have lower GWP over the life cycle (module A, B, C), although the advantage of biogenic carbon in module A is neutralized in module C. Figure 3 shows the sum of module A, B and C. From the analyzed buildings, buildings with CLT construction have the lowest emissions over the buildings life cycle. This is attributed to less fossil materials used in CLT buildings than in timber frame

buildings. Timber frame constructions need more additional materials such as metal connection parts, which contribute to fossil GHG emissions. GWP in kg CO₂ eq/m² GEA over lifecycle for buildings with CLT construction is around 160, for timber frame buildings between 182 and 248 kg CO₂ Eq. The analyzed hybrid buildings vary from 214 to 181 if a wooden attic is also built. Mineral buildings have values between 217 and 340 kg CO₂ eq.

2.1.5 Factor F_M for efficient use of wooden material

A factor for efficient use of wooden material in the construction of buildings is introduced to relate GHG emissions to use of wooden material. The comparison is performed on building level to account for the interconnection of different construction methods. The factor for material efficient use of wooden material in (timber) buildings (F_M) is defined as shown in Eq. 1. It connects the input of wooden material with the environmental effects. The amount of possible GHG emission of buildings over the buildings life cycle (module A + B + C) is combined with the amount of renewable material used—shown through the stored carbon in CO₂ per one m² per GEA of the buildings over the building life cycle.

Through that the different construction types are evaluated. Calculations are done over the whole lifecycle of 50 years. According to EN 15978, module D is separately calculated and always shown separately. Equation 1 shows the factor F_M for material efficient use of wooden material in (timber) buildings.

$$F_M = \frac{GWP(\text{module A} + \text{B} + \text{C})[\text{kg CO}_2/\text{m}^2]}{\text{stored carbon (A)}[\text{kg CO}_2/\text{m}^2]} \quad (1)$$

2.1.5.1 Factor F_M for efficient use of wooden material on exemplary buildings For all of the analyzed buildings this factor is calculated with the GWP values from LCA calculations shown in Fig. 2. Results are demonstrated in Table 2.

The higher the factor F_M is, the better is the relation of GHG emissions to carbon storage, which means: the lower the GHG emissions and the higher (negative) the carbon storage over a buildings lifecycle, the better the material efficiency for the use of wood.

CLT buildings (T_1 and T_2) achieve the highest factors from −0.9 to −0.8. Timber frame buildings (T_3 until T_5) are below with values ranging from −1.9 to −1.0. For comparison: the analyzed mineral buildings reach values of −157.5 up to −9.6 and hybrid buildings range between −9.2 and −1.5. The difference between CLT buildings and timber frame buildings is not distinct. A mix of both structures also reaches a high factor.

The values show that the factor F_M for CLT and timber frame buildings are in the same range compared to mineral

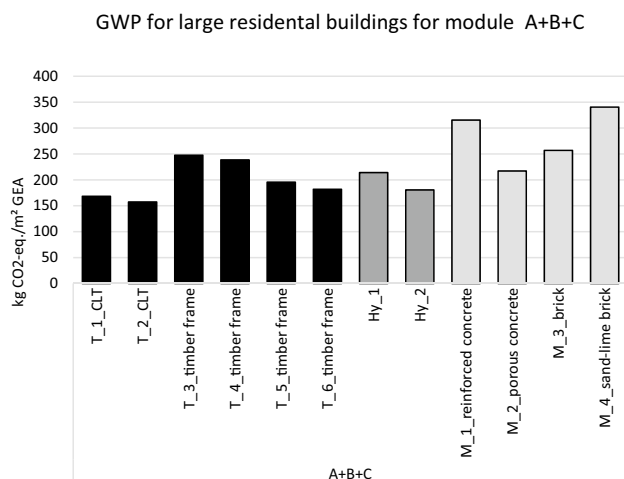


Fig. 3 LCA results for module A+B+C in total, carbon storage is zero

Table 2 Calculated factors F_M for efficient use of wooden material in the analyzed buildings

Number	Building type	GWP (module A, B, C) (kg CO ₂ eq/m ² GEA)	Stored carbon (over 50 years) (kg CO ₂ eq/m ² GEA)	F_M
T_1	Timber	168.3	– 173.8	– 0.97
T_2	Timber	157.4	– 209.8	– 0.75
T_3	Timber	247.6	– 133.5	– 1.86
T_4	Timber	238.7	– 180.7	– 1.32
T_5	Timber	195.6	– 109.4	– 1.79
T_6	Timber	181.9	– 185.8	– 0.98
Hy_1	Hybrid	214.1	– 23.3	– 9.18
Hy_2	Hybrid	180.7	– 118.8	– 1.52
M_1	Mineral	315.5	– 2.0	– 157.47
M_2	Mineral	217.2	– 22.6	– 9.61
M_3	Mineral	257.0	– 2.5	– 102.61
M_4	Mineral	340.5	– 11.4	– 29.80

and hybrid buildings. Therefore, in life cycle perspective the factor F_M for large residential buildings can be seen as independent from the type of timber construction on building level. The GHG emissions of module A + B + C are fossil GHG emissions, since the biogenic carbon was charged off the system in module C. The fossil GHG emissions (module A + B + C) for timber buildings are lower than for hybrid or mineral buildings (see Fig. 3) with higher carbon storage over the life cycle. The decision on choice of timber construction (CLT or timber frame) should be made on structural conditions and technical requirements.

2.1.5.2 Uncertainty, open questions Module D allows supplementary information beyond the building life cycle. It contains benefits (GHG emissions with negative sign) and environmental loads (GHG emissions with positive sign) resulting from recycling of materials, reuse of products (material substitution) and recovery of energy (energetic substitution). In module D the GHG-emissions of all building elements are summed up. The module might be an indicator of how much material is available for reuse and secondary use.

The benefits and loads are calculated according to the current level of knowledge about recycling. In fact, the recycling potential can first be used after 50 years when the end of life of the building is reached and in dependence on how people will then deal with building waste. Furthermore, it is uncertain yet if the recycling potential can be completely utilized or if waste categories cannot be separated entirely.

In this study, the energetic substitution scenarios were used for timber products. Material substitution for wooden products is basically possible but the actual number of doing so is rather small. Höglmeier et al. (2013) stated that 25% of recovered wood is in condition for reuse and additional 21% is channeled in other secondary application.

Benefits can be generated through:

- energy recovery from materials with calorific value (e.g. wooden materials, plastics)
- material recycling from building materials for recycling (e.g. concrete)
- material recycling from metals for recycling

The recycling options are chosen as indicated in the oekobau.dat.

Figure 4 shows the results for module D. All buildings have negative GHG emissions, which means that benefits can be achieved with timber and mineral construction methods. Still the benefits of timber buildings are higher due to the masses of wood which are thermally recovered. High quantity of renewable material therefore results in large benefits for energetic or material substitution of raw material.

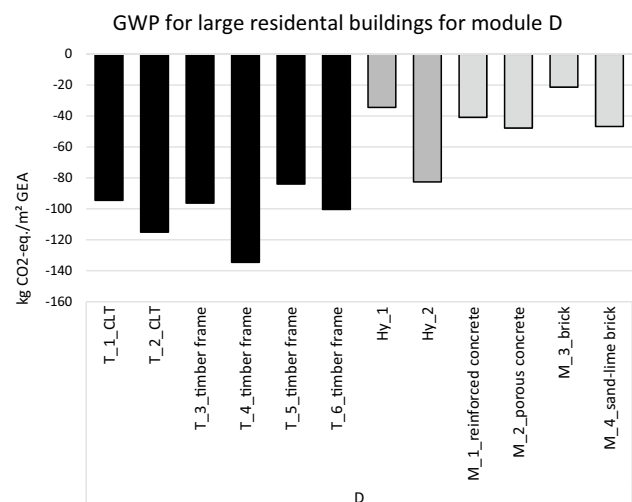


Fig. 4 GWP for module D with different construction types (timber, hybrid, massive)

2.2 Building element level

2.2.1 Optimizing building elements

Results for the factor on building level include the whole building. To influence the factor optimization must be done on the level of building elements. Here, the main building elements are exterior walls, floors/ceilings, roofs and load-bearing internal walls. In timber buildings, these building elements can be constructed in different ways, which then influence the use of various building products.

A building can be divided both in terms of structural elements and according to its functionalities. Each building element must comply with different functionalities, which are possible to achieve through a variety of constructions. The choice for timber construction mainly stems from knowledge, frame conditions, cost, and optimum performance of necessary requirements. In this section, the building elements are discussed separately. For the purpose of this research question, only timber constructions are highlighted

to discuss advantages of timber frame construction and massive timber construction (mainly realized in CLT). All mineral constructions are left aside.

2.2.1.1 Exterior wall Two exemplary constructions from the calculated projects are shown in Fig. 5. The function of the exterior wall is to comply with Energy Conservation Act (EnEV 2014) and therefore to realize specific standard for U-value. Additionally, a certain degree of heat insulation for summer days needs to be reached. The thickness of the exterior wall depends on these parameters. In timber frame construction, insulation can be placed in the same layer as structural elements, whereas in CLT construction, insulation and structural elements are two separate layers. This gives frame construction an advantage in terms of less thickness and therefore, in space-restricted areas, it can achieve slightly more interior space with the same thermal insulation. Additionally, prefabrication of whole elements including all layers (e.g. windows) is more frequently realized in timber frame construction (le Roux and Ott 2014). On a whole, CLT construction is transported from the factory directly to the building site and all additional layers are added on site.

2.2.1.2 Floors/ceiling For floors, sound insulation is a vital issue. Here, filling can be placed in between the beam in timber beam construction. In contrast, CLT construction has the advantage of slightly more mass and therefore slightly lower construction height, see Fig. 6 and detailing (Gräfe et al. 2015). Additionally, this construction allows for combined wood-concrete floor constructions. Depending on fire safety regulations, CLT can remain visible for ceiling for design reasons.

2.2.1.3 Roof In roofs with rafters, at least part of the insulation can be placed between them. This can save valuable height when roof ridge heights are restricted. In contrast, CLT construction is a planar structure, allowing for the realization of free forms.

2.2.1.4 Load-bearing internal wall and party walls Due to the planar structure, thinner wall thickness can be realized

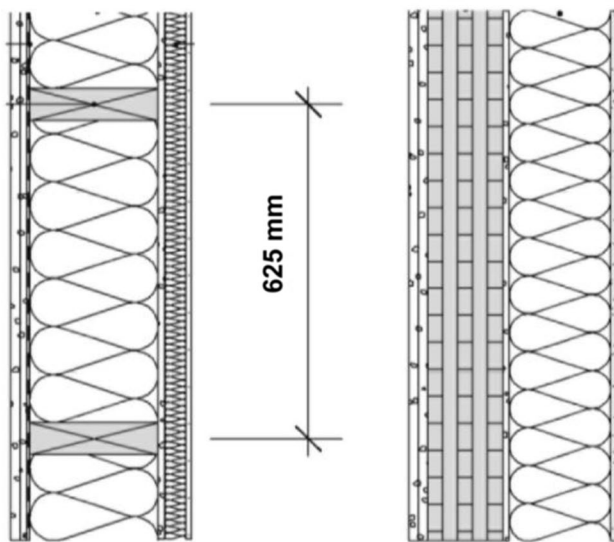


Fig. 5 Exemplary detailing for exterior walls (horizontal section): timber frame construction (left) and CLT construction (right) (Gräfe et al. 2015)

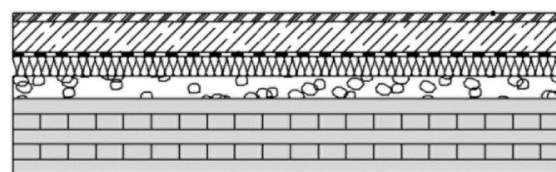
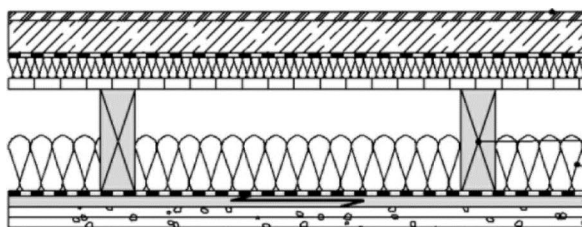


Fig. 6 Exemplary detailing for floors: timber frame construction (left) and CLT construction (right) (Gräfe et al. 2015)

with CLT construction. Depending on construction and on fire safety regulations, it is possible to leave building elements without covering (gypsum board), thus, allowing for visible timber walls.

2.2.2 Interrelations between aspects

Overall, timber frame construction retains advantages where construction is restricted due to external limits to space or height. CLT construction can, however, also be used (when in compliance with fire safety regulations) with less additional gypsum board, thus preserving the wooden surface, and realizing heat and humidity storage effects. In this section, therefore, the technical features of timber frame construction and CLT construction shall be generally described. The choice for one or the other construction method often depends on which one offers easier implementation of requirements for fire safety, sound protection, thermal comfort, load-bearing structure, and, additionally, future recyclability and reuse.

2.2.2.1 Load-bearing structure Suitable materials need to be selected according to their purpose on a case-by-case basis.

From an environmental perspective, the structural frame has a far greater impact than inner components and surface material (Takano et al. 2014a). Advantages of CLT construction in terms of structural elements are: (1) high flexibility in load transfer; (2) possibility for thinner wall construction due to planar load transfer; (3) load-bearing structure for floors can be biaxial; (4) depending on detailing, bracing can be realized even without further covering/sheeting; (5) small openings for building equipment can be made in the structure without the need for detailed calculations. Advantages of timber frame construction are (1) lower timber material consumption to achieve (nearly) same structural performance; (2) sheeting for fire safety reasons can be used for bracing; (3) different requirements can be realized in the same layer, therefore thinner construction thickness is possible.

The load-bearing structure and the exterior wall can be built using two separate elements. This advantage is used by hybrid buildings (Neubauer-Letsch et al. 2013), which have a load-bearing structure in mineral construction (mainly concrete or steel) and then are constructed with timber frame exterior walls. Timber buildings also make use of this structure and use CLT for load-bearing structure and timber frame construction for exterior walls.

2.2.2.2 Sound insulation, thermal insulation, thermal comfort Sound protection is an important issue for timber construction. On the one hand, load-bearing construction from CLT has higher mass. The height can be smaller in regard to

total setup (for a detailed catalog see Gräfe et al. 2015). On the other hand, in timber beam construction the filling can be placed between the beams, as has been the case in timber beam constructions over the past centuries.

In terms of thermal insulation, the most important issue is the thickness of the insulation layer. Here space in-between timber frame construction can be used as insulation layer and additional layers can be applied where necessary. Wall mounting and detailing with all the important construction layers are widely known and also additionally used for renovation projects (Heikkinen et al. 2010). For protection against heat in summer, mass in the building construction is important. This can be achieved through different detailing, for example, part of the construction in mineral construction or visible floor screed. In regard to this issue, hybrid buildings are especially advantageous.

In terms of thermal comfort, visible wooden surface can buffer moisture and heat to a certain extent. This is not connected to any special construction but is realized without problems in CLT construction. For example, Nore and Ollson (2014) show that wood can absorb the high air humidity in bathrooms after showering and, through this, release latent energy into the room.

2.2.2.3 Fire protection Fire protection is one of the most critical issues to be dealt with in (large) timber buildings. Comprehensive planning and precise detailing are mandatory. In terms of considerations regarding different timber constructions, CLT has the advantage of having less hollow space and therefore fewer weaknesses regarding smoke-tight construction. Additionally, in case of fire, insulation between beams cannot fall out. Depending on requirements, however, both construction types are equally possible with special detailing. Possible construction details are shown in Gräfe et al. (2015) with enhancement in Hafner (2017).

2.2.2.4 Design From a design perspective, the approach to timber frame construction is through a grid structure. Depending on the site, the transportation size or crane loads are also included in the design process. With CLT construction, extraordinary forms can be realized and large openings can be implemented relatively easy through girders. Additionally, in terms of design purposes, the freedom of form is greater with CLT construction. For construction in existing building stock and retrofit solutions, timber frame construction is especially wise (Heikkinen et al. 2010). Generally speaking, a minimum of knowledge of timber detailing is necessary for designing with this material as shown in Hafner (2017).

2.2.2.5 Reuse, adaption of buildings and end-of life options The options for reuse and in general end-of-life strategies depend strongly on the possibility of disassem-

bling the construction. The different layers of gypsum board, timber frame, and insulation need to be separated. Due to the extensive use of screws, proper disassembly is rarely done. CLT constructions are connected using fewer fasteners. Here, the amount of plane wooden material is relatively high. Under normal circumstances, this material can be separated correctly. For example, in floors the additional sound insulation layers are only loosely placed on the floor. A separation layer is placed underneath the wet screed so that disassembly can be easily achieved without material mix. Large pieces of plane timber can then be reused with some modifications in size and reassembled for a new purpose.

Regarding the long-term durability of buildings, the possibility of adapting buildings for changing users and functionalities is important. As plans often do not exist after several decades of usage, the easier construction can be adapted, the more flexible the building is in terms of the further use of the building. Today, it is argued that urban densification is easily achieved using timber construction, as only moderate loads are added. Perceptively, thoughts must be given to how density can be achieved through adding stories to timber buildings. This is possible in CLT construction, because the thickness of the walls is not normally calculated based on maximal load, but is calculated according to fire protection regulations and sound insulation, etc.

3 Results and discussion

3.1 Results

The results from Sect. 2 can verify the results shown in various studies before. There are lower GWP for timber buildings than for equivalent mineral buildings. Only the amount of reduction varies as the results in the article are calculated without potential benefits (module D) shown in Fig. 4. Even if the difference between timber and mineral buildings is small it must be noted that this is per m² GEA for one building. As GEA of large buildings in these calculations vary between 700 m² and more than 6000 m² the possible reductions add up.

The results show that there are variations in GWP in timber buildings depending mainly on the load-bearing construction. If this construction is realized in CLT lowest GWP can be reached, whereas the range of timber frame construction depends on additional structural elements made in wood. As each building must meet specific targets, only general tendencies can be pointed out regarding the choice of construction. In terms of material efficient use of wood, timber frame construction is preferable. In terms of the amount of temporary carbon storage, CLT construction is better. Thus, decisions on construction can only be made

based on compliance with all requirements. Exterior wall in lightweight timber frame construction not only saves space, it is easily prefabricated and assembly is fast. Load-bearing structure in CLT holds advantages in structural elements, high carbon storage, as well as advantages in requirements for fire safety, sound insulation, and offers a wide range of design options for making the timber structure visible.

As Takano et al. (2014a) have shown, CLT has high carbon storage and large quantity of PERM but is economically cost intensive in the beginning and has high primary energy consumption (PERE + PENRE). Laminated veneer lumber (LVL) is seen here as a well-balanced option, exhibiting efficient material consumption, while still retaining reasonable carbon storage compared to mineral construction. When it comes to decision-making, it is thus important to understand these characteristics. Specific decisions should be made after considering the pros and cons of each choice based on necessary requirements to be fulfilled.

3.2 Discussion

Timber frame buildings can generate large volumes of space and load-carrying capacity with modest amounts of material (light weight). Historically, it was developed to build large volumes (studwork houses and barns) with limited amounts of material and without the use of additional building equipment such as cranes. As early as the medieval period, due to wood shortages especially of large dimensions, half-timber construction was prevalent in most of Central Europe (Suzuki 1979). Since requirements on fire safety, sound insulation, and building physics are dominant for construction elements and are more easily realized with heavy constructions, timber construction with CLT construction proves most suitable. The area of conflicts lies between load-bearing capacity on the one side and building physics, sound insulation, and fire safety on the other side. Today, dimensioning and detailing is mainly based on building physics and not on load-carrying capacity. The possibility of fulfilling both functions—load-bearing capacity and building physics—is one advantage of timber construction compared to mineral construction.

Overall, a more precise definition of material efficiency in timber buildings could prove very useful for future discussion. On the one hand, less consumption of wooden material for a construction is one factor, which then leads to mixed product (e.g. Fadai and Winter 2015). On the other hand, CLT construction could also be considered as material efficient use of wood, for example when the product leads to less use of other materials (for example, gypsum) and the higher possibility of reuse before timber cascading is applied. Moreover, this option accompanies higher carbon storage. Thus, there is a clear demand for further studies to

work out how material efficiency in timber buildings should be defined, or if it is preferable to keep it open.

A factor for efficient use of wooden material can help to show material efficiency for timber buildings. However, optimization must be made on the level of building elements. The developed environmental factor F_M then is only one of many indicators which can support the decision making process for specific construction types. The analyzed large residential timber buildings achieve factors from -1.9 to -0.8 . This corridor is not representative up to now and further research with more sample buildings is required to attain authoritative values which then could be used as input requirement in the planning process of timber buildings. This article is focused on large residential buildings; small residential buildings and non-residential buildings must be investigated separately. The factor F_M depends significantly on the amount of wood which is used in the building's construction and the amount of additional non-renewable building products in the construction. Due to different building regulations, for example fire protection requirements for specific building sizes, the factors F_M will differ in dependence of the analyzed building types. When calculating the factor F_M it is preconditioned that the buildings to be compared are functionally equivalent and that the LCAs are performed according to the actual standards EN 15978, EN 15804 and EN 16485.

This article shows that it is important to look at theoretical results from LCA calculations together with practical implementation in buildings. The factor hereby shows that there are no unambiguous advantages for either CLT or timber frame construction. However, the factor shows that building with wood is a realistic opportunity to reduce GWP in buildings.

4 Conclusion

The building sector has been identified as a major player in reducing GHG emissions in Europe.

From an environmental perspective, wooden products have advantages compared to mineral products for the building sector, like lower GWP over life cycle and additional temporary carbon storage. An approach that promotes timber buildings can therefore contribute to reaching climate protection targets. All timber buildings must be based on the requirement that the wood derives from sustainable forestry practices. The positive factor of carbon storage and lower GHG emissions in timber building products compared to mineral products should always be studied with material efficiency in mind. Due to stronger competition on wood material, both wood material efficiency in timber products and carbon storage capacity should always be considered in the decision-making process.

It was shown that it is not enough to only focus on one of the issues, as both are legitimate and important aspects from an environmental perspective. Case-specific studies are necessary to find the best solution that fulfills material efficient use of wood and contributes to reaching climate protection goals through temporary carbon storage and lower GWP. From an environmental perspective, timber constructions and buildings need to be further optimized to meet both targets.

Timber construction must fulfill various requirements. Both timber frame construction and CLT construction have their specific advantages for certain building parts. The decision which construction is used needs to be made on project-by-project basis. For larger (residential) buildings the typology of hybrid buildings can generally be seen as a forward-looking construction. Hybrid buildings incorporate carbon storage through the external walls. The load-bearing construction in prefabricated CLT is interesting, especially in terms of sustainable urban re-densification and raising the heights of buildings. A potentially significant development—CLT construction in beech—still is under further study, which could possibly allow for combining high load-carrying capacity with lower material consumption. Additionally, reusing timber constructions for a second lifetime needs further research.

Acknowledgements The named research project “Greenhouse gas balances for timber buildings—Implementation of new requirements for life-cycle-assessments and calculation of empiric substitution factors (GHG—timber buildings)” was funded by the Waldklimafonds, a joined research grant of the Federal Ministry of Food and Agriculture and the Federal Ministry for the Environment, Nature Conservation, Building and Nuclear Safety. The authors gratefully acknowledge the assistance of all concerned.

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